ALLIANCE UNIVERSITY · BCA (AI/ML) · DBMS ASSIGNMENT SERIES

05

ASSIGNMENT 5

Database Creation and Table Implementation

Online Food Delivery System

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IN THIS ASSIGNMENT

01 Create schema

02 Define keys

03 Connect tables

04 Execute DDL

A consistent relational design carried forward from Assignments 1 and 2.

1 Objective and Brief Coverage

This assignment turns the normalised design from Assignment 3 into a working **MySQL** database: one database, eight InnoDB tables, nine foreign-key relationships and the indexes needed for common lookups. Value rules (UNIQUE, CHECK, DEFAULT) are added in Assignment 6.

1	8	9	2
database	tables	foreign keys	extra indexes

	Brief requirement	Where it is covered
✓	**Create the database**	Section 5: CREATE DATABASE online_food_delivery
✓	**Implement all tables with SQL DDL**	Section 5: eight CREATE TABLE statements; Section 3: data dictionary
✓	**Tables and relationships properly defined**	Section 2 (ER diagram), Section 4 (relationship table), foreign keys in the DDL

2 ER Diagram

Order_Item resolves the many-to-many relationship between Orders and Menu_Item. Order totals are derived from line items rather than stored twice. The link from Orders to Address is a **composite** foreign key (Address_ID, Customer_ID), which guarantees that an order uses an address owned by the ordering customer.

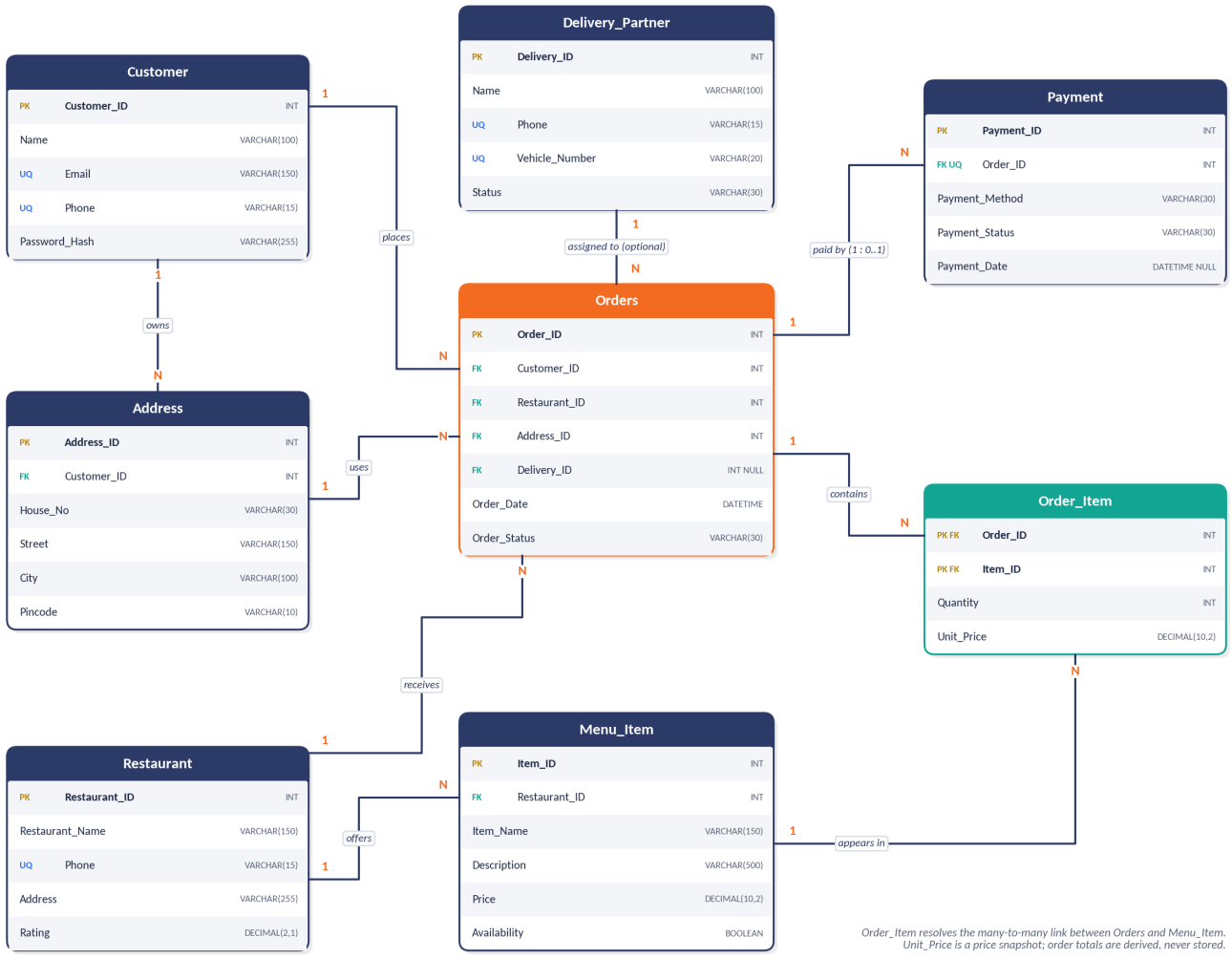


Figure 1. Relational ER diagram of the eight tables. 1 = parent side, N = child side.

3 Data Dictionary

Table	Column	Type	Null	Keys / rules	Note
Customer	Customer_ID	INT	NO	PK, AUTO_INCREMENT	Surrogate identifier
	Name	VARCHAR(100)	NO		
	Email	VARCHAR(150)	NO	UNIQUE (A6)	Login / contact
	Phone	VARCHAR(15)	NO	UNIQUE (A6)	Text, to keep leading zeros and +
	Password_Hash	VARCHAR(255)	NO		Hash only, never plaintext
Address	Address_ID	INT	NO	PK, AUTO_INCREMENT	
	Customer_ID	INT	NO	FK, part of UNIQUE (Address_ID, Customer_ID)	Owner of the address
	House_No	VARCHAR(30)	YES		Optional
	Street	VARCHAR(150)	NO		
	City	VARCHAR(100)	NO		
	Pincode	VARCHAR(10)	NO		Text: postal codes are not numbers

Table	Column	Type	Null	Keys / rules	Note
Restaurant	Restaurant_ID	INT	NO	PK, AUTO_INCREMENT	
	Restaurant_Name	VARCHAR(150)	NO		
	Phone	VARCHAR(15)	YES	UNIQUE (A6)	
	Address	VARCHAR(255)	NO		Single line of text
	Rating	DECIMAL(2,1)	NO	CHECK 0 to 5 (A6)	One decimal, e.g. 4.5
Menu_Item	Item_ID	INT	NO	PK, AUTO_INCREMENT	
	Restaurant_ID	INT	NO	FK	
	Item_Name	VARCHAR(150)	NO		
	Description	VARCHAR(500)	YES		
	Price	DECIMAL(10,2)	NO	CHECK >= 0 (A6)	Exact money type
	Availability	BOOLEAN	NO	DEFAULT TRUE (A6)	Alias of TINYINT(1)
Delivery_Partner	Delivery_ID	INT	NO	PK, AUTO_INCREMENT	
	Name	VARCHAR(100)	NO		
	Phone	VARCHAR(15)	NO	UNIQUE (A6)	
	Vehicle_Number	VARCHAR(20)	NO	UNIQUE (A6)	
	Status	VARCHAR(30)	NO	CHECK list (A6)	Available / Busy / Offline
Orders	Order_ID	INT	NO	PK, AUTO_INCREMENT	
	Customer_ID	INT	NO	FK + composite FK	
	Restaurant_ID	INT	NO	FK	One restaurant per order
	Address_ID	INT	NO	composite FK with Customer_ID	
	Delivery_ID	INT	YES	FK, ON DELETE SET NULL	NULL until a partner is assigned
	Order_Date	DATETIME	NO	INDEX; DEFAULT now (A6)	
	Order_Status	VARCHAR(30)	NO	INDEX; CHECK list (A6)	Life-cycle state
Order_Item	Order_ID	INT	NO	PK (1/2), FK	Cascades with its order
	Item_ID	INT	NO	PK (2/2), FK	
	Quantity	INT	NO	CHECK > 0 (A6)	
	Unit_Price	DECIMAL(10,2)	NO	CHECK >= 0 (A6)	Price snapshot at order time
Payment	Payment_ID	INT	NO	PK, AUTO_INCREMENT	
	Order_ID	INT	NO	FK, UNIQUE (A6)	At most one payment per order
	Payment_Method	VARCHAR(30)	NO	CHECK list (A6)	
	Payment_Status	VARCHAR(30)	NO	CHECK list (A6)	
	Payment_Date	DATETIME	YES		NULL while unpaid

Rules marked (A6) are added in Assignment 6. Money uses DECIMAL, never FLOAT, so that arithmetic is exact.

4 Relationships

Parent	Child	Cardinality	Implementation
Customer	Address	1 : N	Address.Customer_ID
Customer	Orders	1 : N	Orders.Customer_ID
Address	Orders	1 : N	(Address_ID, Customer_ID) composite
Restaurant	Menu_Item	1 : N	Menu_Item.Restaurant_ID
Restaurant	Orders	1 : N	Orders.Restaurant_ID

Parent	Child	Cardinality	Implementation
Delivery_Partner	Orders	1 : N (optional)	Orders.Delivery_ID may be NULL
Orders	Order_Item	1 : N	Order_Item.Order_ID; lines cascade
Menu_Item	Order_Item	1 : N	Order_Item.Item_ID
Orders	Payment	1 : 0..1	Payment.Order_ID, UNIQUE in A6

Many-to-many

An order contains many menu items and a menu item appears on many orders. The associative table **Order_Item** (key = Order_ID + Item_ID) resolves this and carries the attributes of the relationship: Quantity and Unit_Price.

5 MySQL DDL

Run on an empty schema in MySQL Workbench. Tables are created parents first so that every foreign key has a target.

```
CREATE DATABASE IF NOT EXISTS online_food_delivery;
USE online_food_delivery;
CREATE TABLE Customer (
    Customer_ID INT NOT NULL AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(150) NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    Password_Hash VARCHAR(255) NOT NULL,
    PRIMARY KEY (Customer_ID)
) ENGINE=InnoDB;
CREATE TABLE Address (
    Address_ID INT NOT NULL AUTO_INCREMENT,
    Customer_ID INT NOT NULL,
    House_No VARCHAR(30),
    Street VARCHAR(150) NOT NULL,
    City VARCHAR(100) NOT NULL,
    Pincode VARCHAR(10) NOT NULL,
    PRIMARY KEY (Address_ID),
    -- This composite key lets Orders enforce that the chosen address belongs to the same customer.
    UNIQUE KEY uq_address_owner (Address_ID, Customer_ID),
    CONSTRAINT fk_address_customer FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;
CREATE TABLE Restaurant (
    Restaurant_ID INT NOT NULL AUTO_INCREMENT,
    Restaurant_Name VARCHAR(150) NOT NULL,
    Phone VARCHAR(15),
    Address VARCHAR(255) NOT NULL,
    Rating DECIMAL(2,1) NOT NULL,
    PRIMARY KEY (Restaurant_ID)
) ENGINE=InnoDB;
CREATE TABLE Menu_Item (
    Item_ID INT NOT NULL AUTO_INCREMENT,
    Restaurant_ID INT NOT NULL,
    Item_Name VARCHAR(150) NOT NULL,
    Description VARCHAR(500),
    Price DECIMAL(10,2) NOT NULL,
    Availability BOOLEAN NOT NULL,
    PRIMARY KEY (Item_ID),
    CONSTRAINT fk_menu_restaurant FOREIGN KEY (Restaurant_ID)
        REFERENCES Restaurant(Restaurant_ID) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;
CREATE TABLE Delivery_Partner (
    Delivery_ID INT NOT NULL AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15) NOT NULL,
    Vehicle_Number VARCHAR(20) NOT NULL,
    Status VARCHAR(30) NOT NULL,
    PRIMARY KEY (Delivery_ID)
) ENGINE=InnoDB;
```

```
CREATE TABLE Orders (
    Order_ID INT NOT NULL AUTO_INCREMENT,
    Customer_ID INT NOT NULL,
    Restaurant_ID INT NOT NULL,
    Address_ID INT NOT NULL,
    Delivery_ID INT NULL,
    Order_Date DATETIME NOT NULL,
    Order_Status VARCHAR(30) NOT NULL,
    PRIMARY KEY (Order_ID),
    KEY idx_orders_date (Order_Date),
    KEY idx_orders_status (Order_Status),
    CONSTRAINT fk_orders_customer FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_orders_restaurant FOREIGN KEY (Restaurant_ID)
        REFERENCES Restaurant(Restaurant_ID) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_orders_address_owner FOREIGN KEY (Address_ID, Customer_ID)
        REFERENCES Address(Address_ID, Customer_ID) ON UPDATE CASCADE ON DELETE RESTRICT,
    CONSTRAINT fk_orders_delivery FOREIGN KEY (Delivery_ID)
        REFERENCES Delivery_Partner(Delivery_ID) ON UPDATE CASCADE ON DELETE SET NULL
) ENGINE=InnoDB;
CREATE TABLE Order_Item (
    Order_ID INT NOT NULL,
    Item_ID INT NOT NULL,
    Quantity INT NOT NULL,
    Unit_Price DECIMAL(10,2) NOT NULL,
    PRIMARY KEY (Order_ID, Item_ID),
    CONSTRAINT fk_order_item_order FOREIGN KEY (Order_ID)
        REFERENCES Orders(Order_ID) ON UPDATE CASCADE ON DELETE CASCADE,
    CONSTRAINT fk_order_item_menu FOREIGN KEY (Item_ID)
        REFERENCES Menu_Item(Item_ID) ON UPDATE CASCADE ON DELETE RESTRICT
) ENGINE=InnoDB;
CREATE TABLE Payment (
    Payment_ID INT NOT NULL AUTO_INCREMENT,
    Order_ID INT NOT NULL,
    Payment_Method VARCHAR(30) NOT NULL,
    Payment_Status VARCHAR(30) NOT NULL,
    Payment_Date DATETIME NULL,
    PRIMARY KEY (Payment_ID),
    CONSTRAINT fk_payment_order FOREIGN KEY (Order_ID)
        REFERENCES Orders(Order_ID) ON UPDATE CASCADE ON DELETE CASCADE
) ENGINE=InnoDB;
```

6 Design Decisions

- **InnoDB** is the engine that supports foreign keys and ACID transactions.
- **AUTO_INCREMENT integers** give stable identifiers independent of display names.
- **Composite foreign key** on Orders makes the database verify address ownership instead of trusting the application.
- **Unit_Price snapshot**: changing Menu_Item.Price never rewrites past orders.
- **No stored totals**: Orders.Total_Amount and Payment.Amount were removed; compute SUM(Quantity * Unit_Price). Taxes, fees and discounts are out of scope.
- **Indexes** on Order_Date and Order_Status speed the date and status filters used in Assignments 8 and 9; primary and foreign keys are indexed automatically by InnoDB.
- **Reset script**: to rebuild, drop children first (Payment, Order_Item, Orders, Menu_Item, Address, Delivery_Partner, Restaurant, Customer) or DROP DATABASE online_food_delivery;.

7 Verifying the Structure

Run these after the script to confirm the tables and keys exist. They were not executed in this document; capture Workbench screenshots as evidence.

```
SHOW TABLES;                          -- expect 8 tables
DESCRIBE Orders;                      -- columns, types, keys
SHOW CREATE TABLE Orders;             -- full DDL incl. foreign keys
-- list every foreign key in the schema
SELECT TABLE_NAME, COLUMN_NAME, CONSTRAINT_NAME,
       REFERENCED_TABLE_NAME, REFERENCED_COLUMN_NAME
FROM   information_schema.KEY_COLUMN_USAGE
WHERE  TABLE_SCHEMA = 'online_food_delivery'
  AND  REFERENCED_TABLE_NAME IS NOT NULL
ORDER BY TABLE_NAME, CONSTRAINT_NAME;
```